AIORI Hackathon 3 · Problem Statement PS-035

# Evidence-Based Operator Network Fault Localisation

Ranking probable fault locations from correlated, verifiable evidence

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Design proposal with a controlled-lab evaluation plan

# Abstract & Problem Description

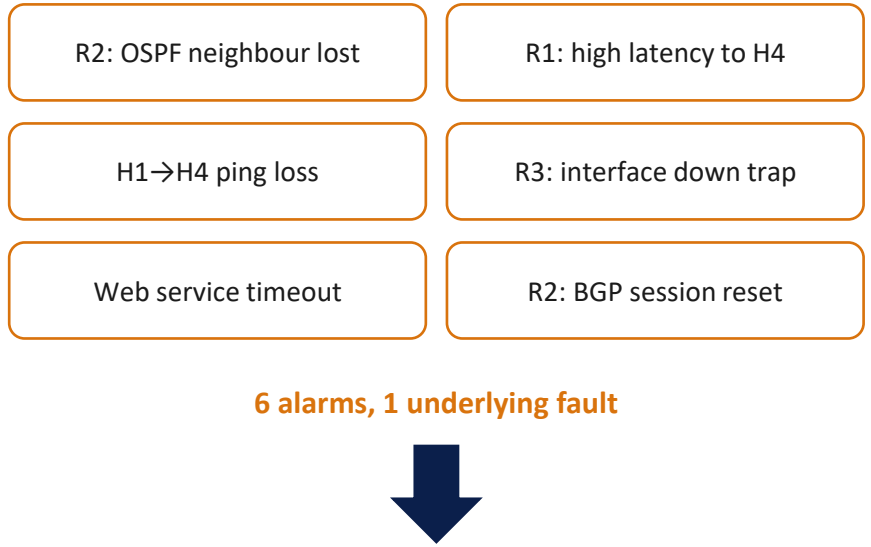
**Abstract.** We propose an evidence-linked fault localiser for operator networks. It fuses SNMP, syslog, active-probe and routing data over a topology model, ranks probable fault locations, separates root causes from downstream symptoms, and reports “inconclusive” when evidence is thin.

## Key problems

- One fault triggers many alarms across devices and layers
- Evidence is scattered: logs, counters, probes, routing state, device CLIs
- Manual correlation is slow, expertise-dependent and hard to audit
- Verdicts without linked evidence or stated uncertainty are not trustworthy

**Goal:** identify probable fault locations from correlated, verifiable evidence.

*Illustrative example*



# Background

Five evidence sources are needed, and each one is partial on its own

| Evidence source | What it tells us | Limitation |
|---|---|---|
| **SNMP polling [1]** | Interface state, error and discard counters | Periodic: short events fall between polls |
| **Syslog events [2]** | Timestamped link, neighbour and protocol events | Vendor wording varies; loss and clock skew |
| **Active probes** | End-to-end reachability, latency, loss | Shows a failing path, not the failing element |
| **Routing state** | Adjacencies and routes seen by each router | Also changes as a consequence of faults |
| **Topology model [4]** | Which nodes, ports and links connect | Wrong if stale or incomplete |

*Sources are grouped by telemetry plane (management, control, forwarding) following the framework in [3].*

## Why diagnosis is hard

- Faults propagate: symptoms appear far from the cause
- A symptom such as “neighbour lost” looks like a cause
- Data is noisy, late and partly missing
- Scope here: an emulated lab, one administrative domain

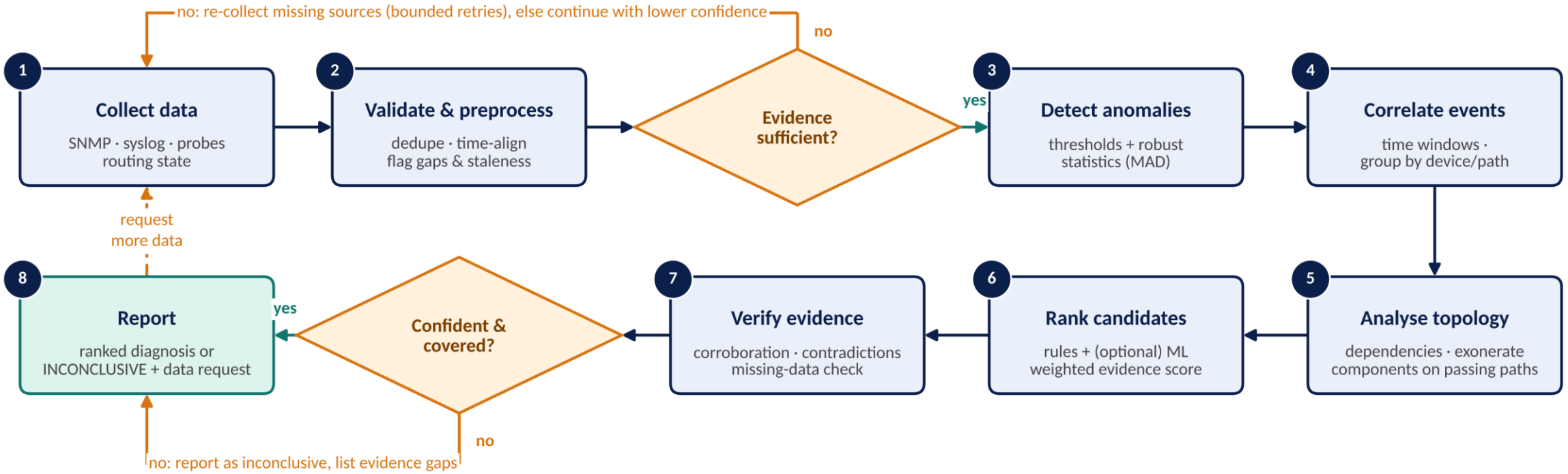
# Literature & RFC Review

Standards define what we collect and model; research shows how to localise faults

| Ref | Source | What it provides | How we use it |
|---|---|---|---|
| **[1]** | RFC 3411 (STD 62), IETF, 2002 | SNMP management architecture | Architecture behind SNMP polling of interface state and counters |
| **[2]** | RFC 5424, IETF, 2009 | Syslog message format and fields | Parsing and normalising event logs |
| **[3]** | RFC 9232, IETF, 2022 | Network telemetry framework and taxonomy | Organising evidence sources and terminology |
| **[4]** | RFC 8345, IETF, 2018 | YANG model for network topologies | Node / port / link concepts for our topology graph |
| **[5]** | Kompella et al., NSDI 2005 | SCORE: risk-model fault localisation across layers | Shared-risk reasoning: find components common to failures |
| **[6]** | Bahl et al., SIGCOMM 2007 | Sherlock: inference graph of multi-level dependencies | Dependency-aware, three-state (up / troubled / down) reasoning |

**Our focus:** a small, transparent localiser for operators, with evidence links, confidence scores and an explicit “inconclusive” state.

# Proposed Solution




### Root cause vs symptom

- Topology graph [4]: alarms downstream of a faulty element are tagged as symptoms
- Failed probe paths implicate shared components [5]; passing paths clear them

### Rules + AI/ML, clear roles

- Rules: direct evidence (link down, neighbour loss)
- Rolling median/MAD: label-free anomaly detection
- Optional ML ranker, kept only if it beats rules on held-out runs

### Ranking and uncertainty

- Score = direct evidence + path coverage + timing + corroboration − contradictions
- Missing telemetry = unknown, not healthy
- Low margin or coverage → “inconclusive”

# Optimization Proposed

From manual evidence hunting to ranked, evidence-linked diagnosis

| Aspect | Before: manual / conventional | After: proposed | Measured by |
|---|---|---|---|
| **Evidence gathering** | Device-by-device log-ins and CLI checks | One time-aligned evidence store | Evidence coverage; time to diagnose |
| **Alarm handling** | Isolated alerts, each investigated alone | Topology-aware correlation into one incident | Top-1 accuracy; false-positive rate |
| **Candidates** | Unranked list of possibilities | Ranked fault candidates with scores | Top-1 / Top-3 accuracy |
| **Explanations** | Unsupported guess or opaque verdict | Every candidate links to timestamped evidence | Share of claims with linked evidence |
| **Uncertainty** | Implicit; confidence unclear | Explicit confidence and “inconclusive” state | Precision, recall, F1 incl. inconclusive |

### How we will measure

- Inject known faults (link down, flap, loss, delay, node down, misconfiguration) at random locations; repeat runs
- Ground truth from the injection log
- Compare: B0 manual checklist, B1 first alarm, B2 rules only, B3 rules + statistics (+ optional ML)
- Metrics: Top-1 / Top-k, precision, recall, F1, mean time to diagnose, false positives on fault-free runs, evidence coverage, processing latency

**No numeric gain is claimed yet:** targets are set after the baseline runs.

# Solution Architecture & Design

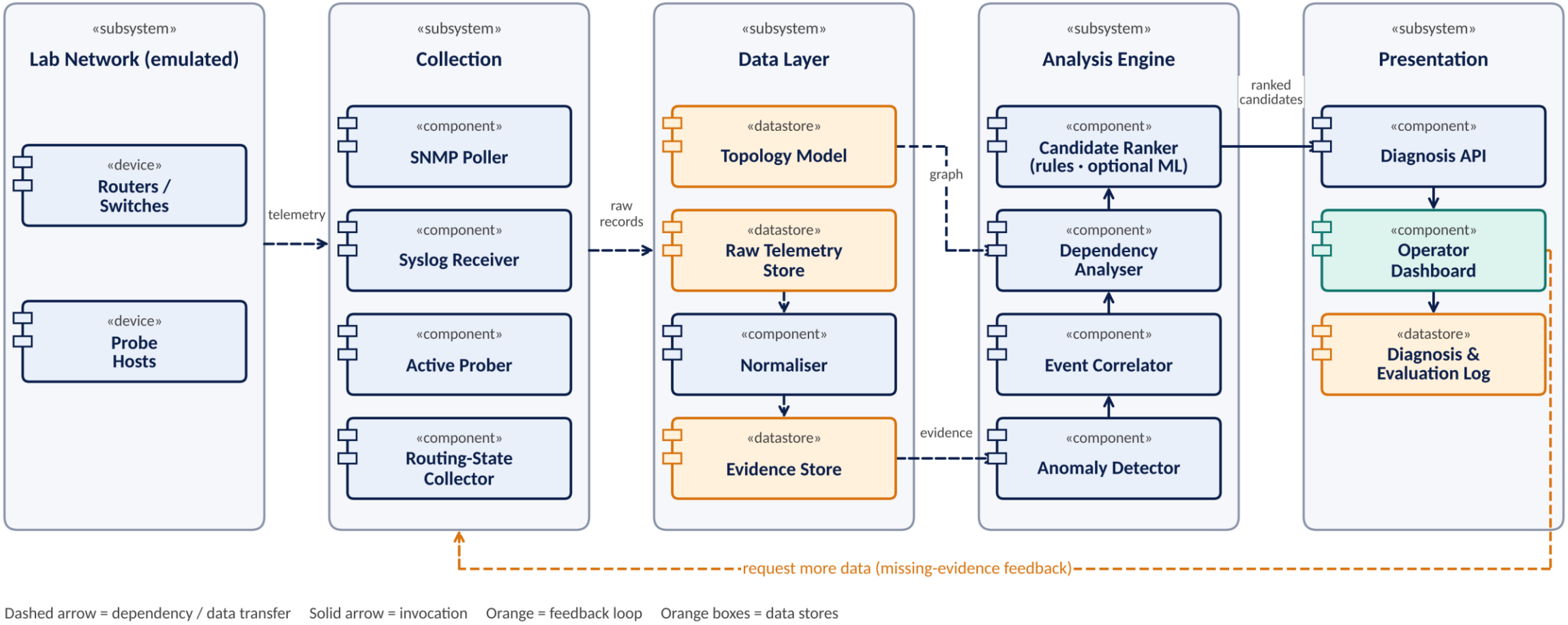
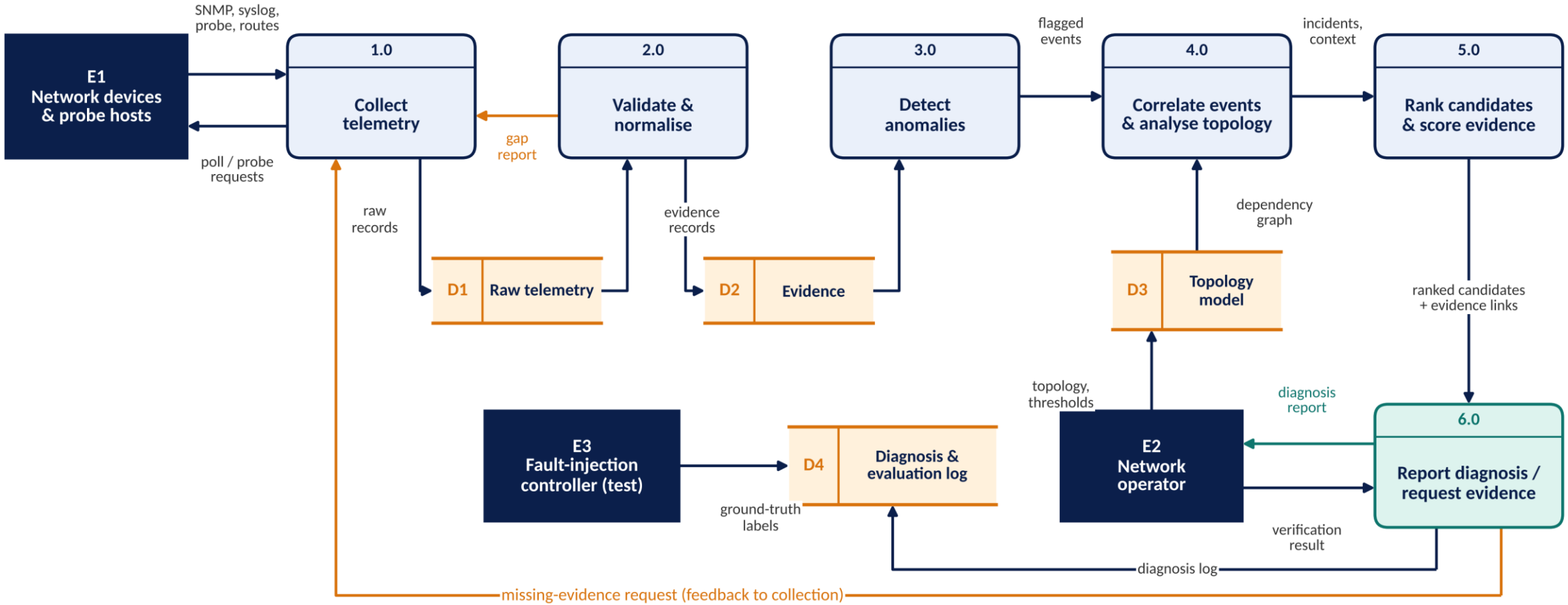



*Figure: System architecture, UML component diagram (proposed design)*

# Data Flow Diagram

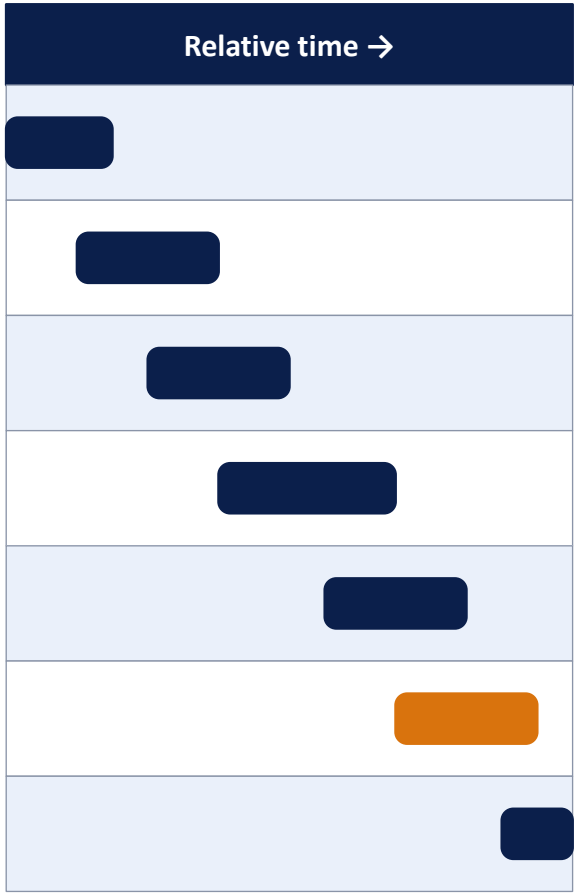



*Level-1 DFD. Processes map to the workflow steps on slide 5: 1.0 = step 1, 2.0 = 2, 3.0 = 3, 4.0 = 4–5, 5.0 = 6–7, 6.0 = 8.*

# Deployment Timeline

Seven phases in relative time units; scale them to the time available

| Phase | Deliverable | Needs | Validation gate |
|---|---|---|---|
| **P1 Lab & scenarios** | Emulated topology; scenario catalogue with ground truth | None | Each scenario replays identically |
| **P2 Telemetry collection** | SNMP, syslog, probe and route collectors | P1 | All sources reach the store during a test fault |
| **P3 Normalise & store** | Evidence schema, stores, topology model | P2 | Replayed logs normalise without loss |
| **P4 Correlate & localise** | Detector, correlator, dependency analyser, ranker | P3 | Rules localise a simple link-down fault |
| **P5 Dashboard & API** | Ranked list, evidence links, confidence, next checks | P3, P4 | Operator walkthrough on 3 scenarios |
| **P6 Test & evaluate** | Repeated fault runs; baselines B0–B3; metrics report | P4, P5 | Metrics computed from logs, not by hand |
| **P7 Integrate & demo** | End-to-end demo with fault-free and missing-data cases | P6 | Live run matches recorded results |




*Scope: one primary fault per incident in this version; multi-fault cases and production-scale validation are future work.*

# Bibliography

**[1]** D. Harrington, R. Presuhn and B. Wijnen, “An Architecture for Describing Simple Network Management Protocol (SNMP) Management Frameworks,” RFC 3411 (STD 62), IETF, Dec. 2002. https://www.rfc-editor.org/info/rfc3411

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**[3]** H. Song, F. Qin, P. Martinez-Julia, L. Ciavaglia and A. Wang, “Network Telemetry Framework,” RFC 9232, IETF, May 2022. https://www.rfc-editor.org/info/rfc9232

**[4]** A. Clemm, J. Medved, R. Varga, N. Bahadur, H. Ananthakrishnan and X. Liu, “A YANG Data Model for Network Topologies,” RFC 8345, IETF, Mar. 2018. https://www.rfc-editor.org/info/rfc8345

**[5]** R. R. Kompella, J. Yates, A. Greenberg and A. C. Snoeren, “IP Fault Localization via Risk Modeling,” in Proc. 2nd USENIX NSDI, Boston, MA, May 2005, pp. 57–70. https://www.usenix.org/conference/nsdi-05/ip-fault-localization-risk-modeling

**[6]** P. Bahl, R. Chandra, A. Greenberg, S. Kandula, D. A. Maltz and M. Zhang, “Towards Highly Reliable Enterprise Network Services via Inference of Multi-level Dependencies,” in Proc. ACM SIGCOMM, Kyoto, Japan, Aug. 2007, pp. 13–24. https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/sherlock_sigcomm_07.pdf

# Thank you for your attention

Ranked fault locations, linked evidence, honest uncertainty.

Questions welcome

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